Lab Report: Training dynamics of deep models

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.01 with the Adam optimizer. The dataset was split into 70 percent training data and the rest for testing.

2 Results

Figure 1 shows the training loss.

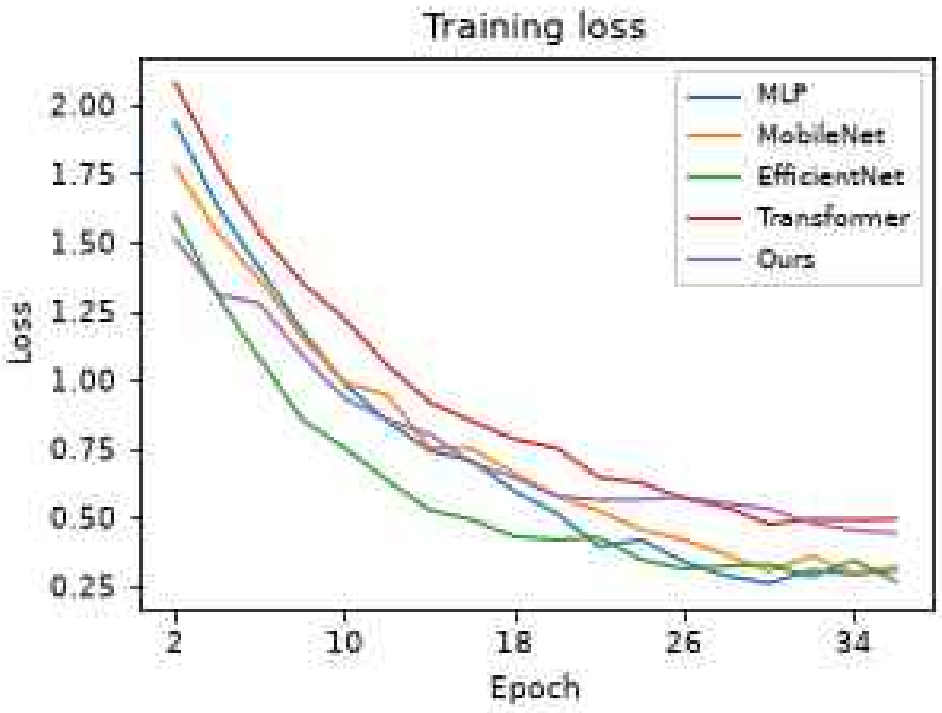


Figure 1: Training loss for the models in this experiment.

At epoch 2, the loss of EfficientNet is 1.598. Transformer has a loss of 2.08 at epoch 2.

Figure 2 shows the test accuracy during training.

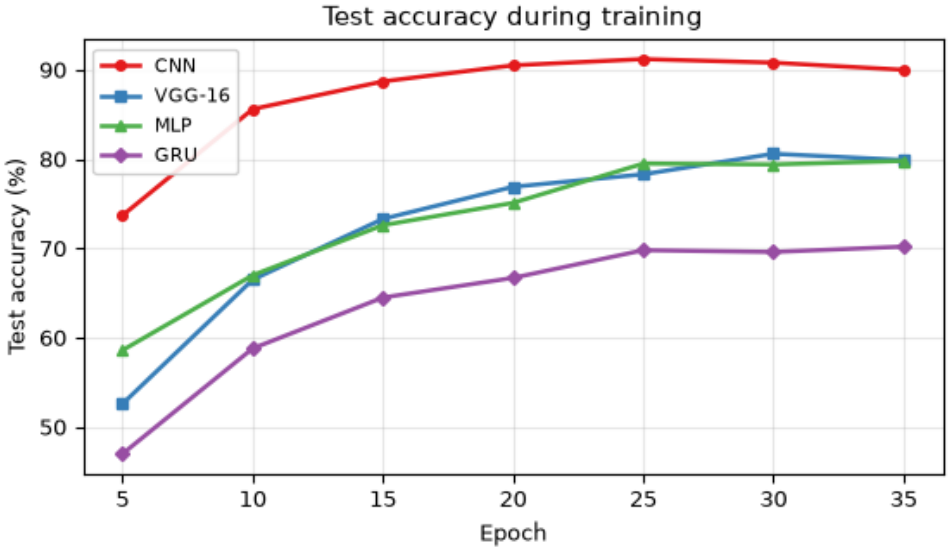


Figure 2: Test accuracy during training for the models in this experiment.

As shown in Figure 2, from epoch 10 to epoch 25, the test accuracy of CNN improves by a relative 6.5%. From epoch 20 to epoch 35, the test accuracy of MLP improves by a relative 6%.

Figure 3 shows the training loss.

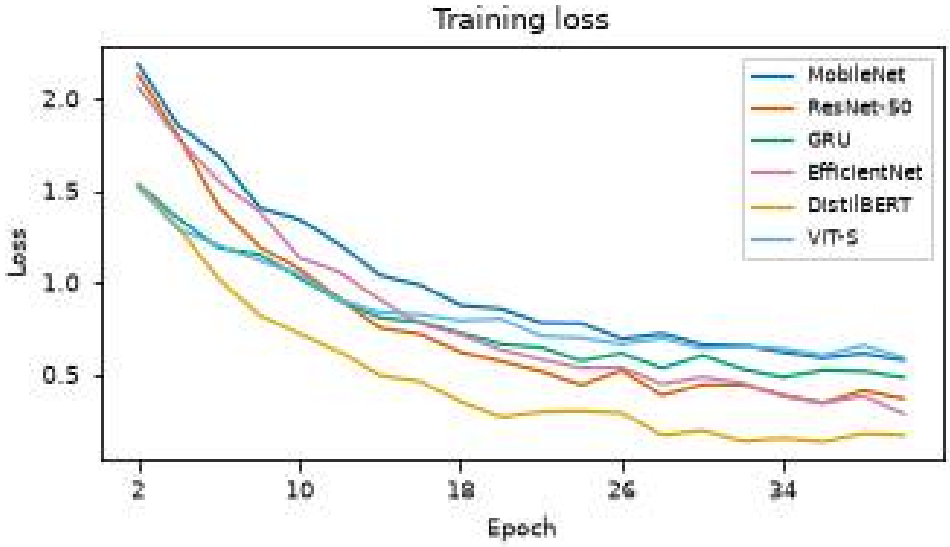


Figure 3: Training loss for the models in this experiment.

At epoch 32, the loss of GRU is 0.302 (Figure 3). ResNet-50 has a loss of 1.514 at epoch 34 (Figure 3). At epoch 26, the loss of ResNet-50 is 0.297 (Figure 3). As shown in Figure 3, at epoch 20, the loss of ViT-S is 0.803.

Figure 4 shows the validation accuracy of ResNet-50 by dataset.

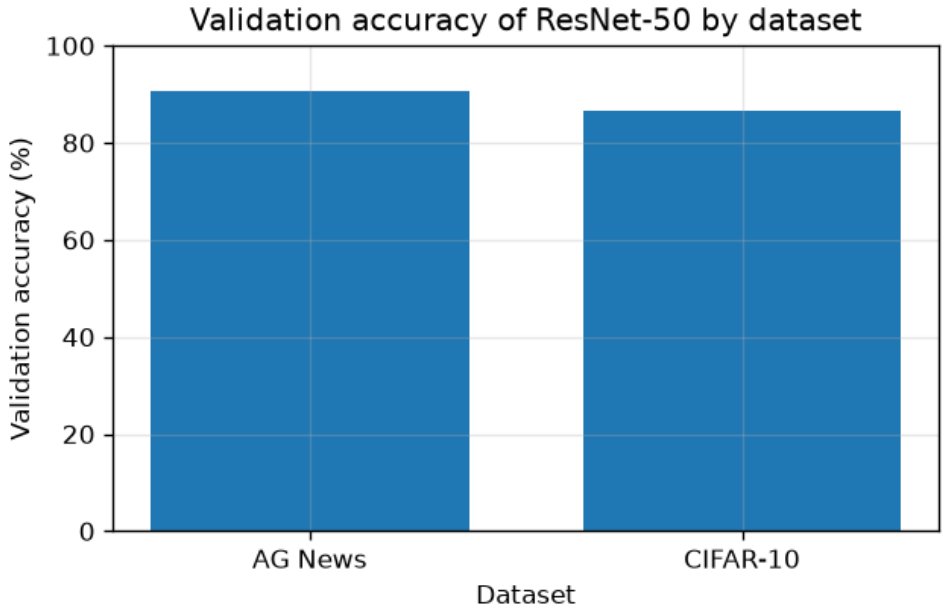


Figure 4: Validation accuracy of ResNet-50 by dataset for the models in this experiment.

ResNet-50 scores about 4 points higher on AG News than on CIFAR-10. ResNet-50 reaches 90.7% validation accuracy on AG News.

3 Discussion

Overall, the results are in line with what we expected from the literature.